

LUCIAN TRUSCA

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DESCRIPTION

Computer science graduate with a background in biotechnology, motivated by projects that turn data into solutions for real-world issues, especially in AI-assisted tooling/software. Experience in cleaning datasets, building dashboards, and applying machine learning and AI methods. I enjoy delivering solutions that are practical and impactful for businesses and end-users.

EDUCATION

MSc in Information Studies – Data Science | University of Amsterdam

Thesis: Exploring Multiomics Data through Priors-Informed Graphical LASSO and Network Diffusion

September 2025 – Ongoing

Amsterdam, The Netherlands

BSc in Computer Science | University of Twente

Thesis: Investigating suitable vision transformer models for wildlife camera trap data

2021 – July 2025

Enschede, The Netherlands

MSc in Molecular Biotechnology | Babes-Bolyai University

Thesis: Bioinformatic characterization of taxonomic diversity in Movile Cave sediment microbiomes

2019 – 2021

Cluj Napoca, Romania

BSc in Industrial Biotechnology | Babes-Bolyai University

Thesis: Optimizing ancient DNA Recovery from archaeological wood samples

2015 – 2019

Cluj Napoca, Romania

TECHNICAL SKILLS

Data & Analytics:

Data cleaning and wrangling (Python/Pandas, SQL, R), exploratory data analysis, visualization (Matplotlib), dashboarding (Dash).

Machine Learning & AI:

Predictive modeling (classical ML), NLP, explainable AI techniques, object detection, time series forecasting.

Programming languages:

Python, R, Java, TypeScript, SQL (PostgreSQL), HTML/CSS.

Frameworks & Libraries:

React, Node.js, Flask, Dash, Pandas, NumPy, Matplotlib, scikit-learn.

Other skills:

Git, Docker, Agile/Scrum, Copilot, Claude Code.

Languages:

English — Native; Romanian — Native; Dutch — A2 (certificate).

SELECTED PROJECTS & EXPERIENCE

Data Analyst for Research Programs – EHA | Data Analysis, Power BI, Excel

- Supported the **annual survey analysis** to demonstrate the impact of EHA Grants and Research Training programs on the professional and personal development of hematology researchers and clinicians.
- Managed data** from the Researchfish platform to track impact through indicators such as collaborations, publications, additional funding, clinical studies, and other parameters.

CS BSc thesis – Meadow bird species detection | Object Detection, Vision Transformers, Docker, Explainable AI, Machine Learning

- Analyzed performance of vision transformer models** for identification of meadow bird species from camera trap images.
- Evaluated model performance** across varying environmental conditions (e.g., day/night) to assess robustness and reliability.
- Applied **explainable AI techniques** to interpret and visualize model behavior, identifying strengths/limitations for wildlife monitoring use cases.

Time series forecasting – Applied Forecasting | Time Series Forecasting, ETS/SARIMAX, Feature Engineering, Python, Model Evaluation

- Built an end-to-end **daily sales forecasting pipeline** for Walmart dataset, predicting **28-day demand**.
- Performed **EDA and preprocessing**.
- Identified **weekly seasonality** (weekend demand peaks), non-stationarity, and time-varying volatility using **rolling statistics, differencing**, and **ACF/PACF** diagnostics.
- Implemented and compared forecasting models: **seasonal naive**, **ETS**, and **SARIMAX with exogenous regressors**.

Security Application Dashboard | Python Dash, Figma, Prototyping, SQL, Docker, Team Collaboration

- Designed, prototyped, and **implemented an interactive dashboard** (Python Dash) for the “DeepCase” command-line tool.
- Designed **interface prototypes** with Figma.
- Gathered weekly **stakeholder feedback** throughout process.
- Collaborated with stakeholders to **define key metrics** and **requirements**.

Cottages – Data science & AI | NLP, Machine Learning, spaCy, GPT 2.0, Python, Data Cleaning

- **Curated** and optimized a **real-world JSON dataset** of cottage listings using Python and NLP techniques to improve data quality and consistency.
- Applied **text preprocessing** with spaCy and other libraries to clean and standardize descriptions.
- Built and evaluated **machine learning models** (logistic regression, decision trees etc) to fill missing values and enhance dataset usability.

Traffic sign detection system – Intelligent Embedded Systems | Python, Embedded Systems, Computer Vision, Arduino, EdgeImpulse

- Developed and deployed a real-time **traffic sign detection** and classification system on an **Arduino**, using the EdgeImpulse platform.
- Engineered a lightweight **image-classification model** achieving over 99% accuracy on 6-class detection in optimal conditions and 75% accuracy on a full 42-class dataset.
- Conducted extensive **field testing** across varied distances, angles, lighting and backgrounds, ensuring robust real-world performance.

Training & management system – Data & Information | HTML, JavaScript (jQuery, Bootstrap), SQL, RESTful API, Agile/Scrum, UI/UX

- Developed a **role-based training management web application** used by administrators, trainers, and trainees.
- Implemented features for **course creation, user management, group training, and trainee progress** tracking.
- Designed **trainee dashboards** with clear views of personal schedules, group memberships, and progress visualizations.

Chessmate Helper – Human Computer Interaction | Python, OpenCV, UI/UX Design (Figma), HCI

- Built an **AI chess assistant** with real-time object detection for move suggestions.
- Conducted **user research** and tested prototypes with players of different skill levels.
- Designed **personas and feedback flows** to guide the user experience.
- Collaborated in a team using **Git** and **Scrum**.

Quality Control Junior Technician – Terapia Cluj | GMP, Analytical Testing, HPLC, UV-Vis Spectrophotometry, Documentation, Pharmaceutical QC

- Performed **quality control testing** of raw materials, in-process samples (January 2021 – July 2021).
- Operated **analytical instruments** including HPLC, UV-Vis spectrophotometry, and pH measurement equipment.
- Maintained accurate **records and laboratory documentation** in alignment with regulatory and quality assurance standards.
- Helped monitor and document **environmental conditions** in controlled production areas, including collecting samples to verify contamination levels met manufacturing standards.

MSc Thesis – Ancient DNA Bioinformatics Pipeline | Bioinformatics, phylogenetics, metagenomics, statistical analysis, microbial community analysis

- **Analyzed cave sediment microbiomes** using metagenomic and transcriptomic data.
- Assessed species abundance and density, applying cluster analysis to **reveal patterns in microbial community** structure.

SELECTED EXTRACURRICULAR ACTIVITIES

Cassini Hackathon (6TH) – International Development & Humanitarian Aid | Machine Learning, Data Analysis, Python, Geospatial Data, Team Collaboration

- Co-developed a **flood prediction solution** in a team setting during a humanitarian-focused hackathon.
- Used **machine learning models** to predict flooding using time-series data.
- Presented end solution to a jury.